Lecture 11, Jan 14, 2026.

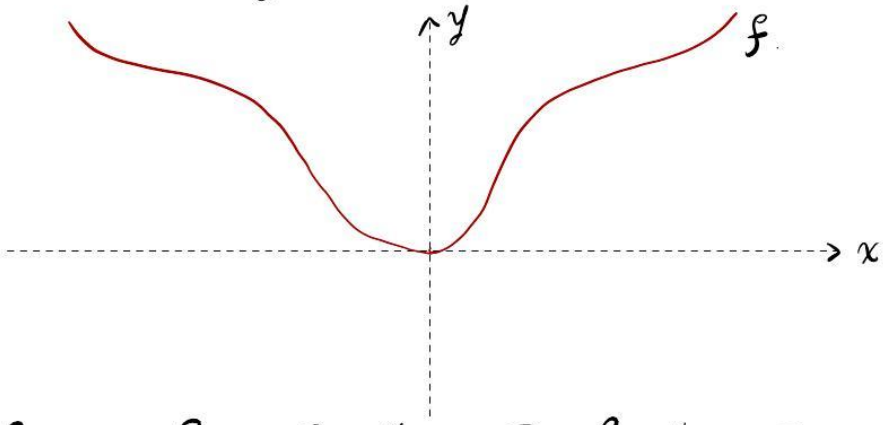
3.5 Guidelines for Sketching a Curve

A). Domain. ($Dom f$)

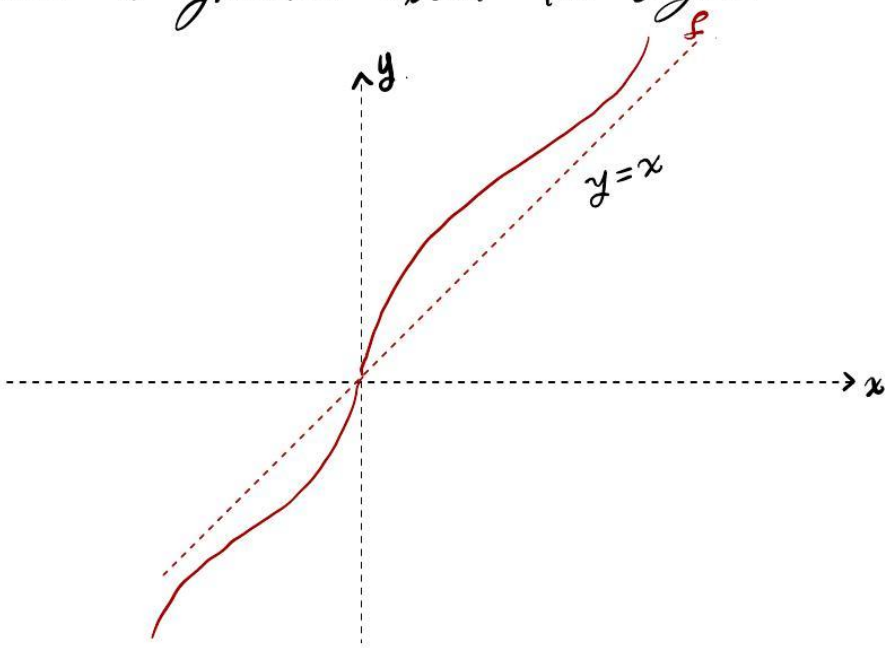
B). Intercepts

C). Symmetry:

i). If $f(-x) = f(x)$ for all $x \in Dom f$ then f is an even function and the curve is symmetric about the y-axis.



ii). if $f(-x) = -f(x)$ for all $x \in Dom f$ then f is an odd function and the curve is symmetric about the origin.



D). Asymtotes

i). Horizontal Asymtotes: if $\lim_{x \to \infty} f(x) = L \Rightarrow y = L$ is HA.

if $\lim_{x \to -\infty} f(x) = K \Rightarrow y = K$ is HA.

ii). Vertical Asymtotes: The line $x=a$ is a vertical asymtote if one of the following statements is true:

$$\lim_{x\to a^+} f(x) = \infty \qquad \lim_{x\to a^+} f(x) = -\infty$$

$$\lim_{x\to a^-} f(x) = \infty \qquad \lim_{x\to -\infty} f(x) = -\infty.$$

iii). Slant Asymtote:

E). Intervals of Increase or Decrease: Use $f'(x)>0$ and solve if inequality.

F). Local Maximums or Minimum values: Find the critical numbers of f where $f'(c)=0$ or $f'(c)$ DNE. Then use the First Derivative test or Second Derivative test to compute these.

G). Concavity and Points of inflection: Compute $f''(x)$ and use

$f''(x)>0$ then $\cup$

$f''(x)<0$ then $\cap$

and inflection points are points s.t $f''(c)=0$ or DNE.

H). Sketch the curve:

Example Use the guidelines to sketch the curve $y = \frac{2x^2}{x^2-1}$.

Sln. A. Domain.

$$x^2-1=0 \iff x=\pm 1 \Rightarrow \text{Dom } f = \mathbb{R}-\{\pm 1\}.$$

B. Intercepts: $x=0 \Rightarrow y=\frac{2(0)^2}{0^2-1} \Rightarrow (0,0)$ y-intercepts.

x-intercepts: $y=0$ then $\frac{2x^2}{x^2-1}=0 \Rightarrow x=0$. $(0,0)$

C). Symmetry: f is even.

$$f(-x) = \frac{2(-x)^2}{(-x^2)-1} = \frac{2x^2}{x^2-1} = f(x).$$

D. Asymptotes: Vertical: the possible Vertical asymptotes are $x = \pm 1$

$$\lim_{x \to 1^+} \frac{2x^2}{x^2-1} = \infty \quad \text{and} \quad \lim_{x \to 1^-} \frac{2x^2}{x^2-1} = -\infty.$$

$$\lim_{x \to -1^+} \frac{2x^2}{x^2-1} = -\infty \quad \text{and} \quad \lim_{x \to -1^-} \frac{2x^2}{x^2-1} = +\infty.$$

So, $x=1$, $x=-1$ are vertical asymptotes.

Horizontal. $\lim_{x \to \infty} \frac{2x^2}{x^2-1} = 2$, $\lim_{x \to -\infty} \frac{2x^2}{x^2-1} = 2$.

So, $y=2$ is a horizontal asymptote.

E. Intervals of Increase / Decrease: $f(x) = \frac{2x^2}{x^2-1}$

$$f'(x) = \frac{4x(x^2-1) - 2x^2(2x)}{(x^2-1)^2} = \frac{4x^3 - 4x - 4x^3}{(x^2-1)^2} = \frac{-4x}{(x^2-1)^2}$$

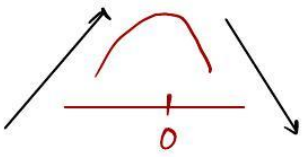
Solving $f'(x) > 0 \iff \frac{-4x}{(x^2-1)^2} > 0 \iff -4x > 0 \iff x < 0$

f is increasing on $(-\infty, -1) \cup (-1, 0)$

and f is decreasing on $(0, 1) \cup (1, +\infty)$.

Critical points: Possible critical points $x=0$, $x=1$, $x=-1$ but since $x = \pm 1 \notin \text{Dom} f$ then the only critical point is $x=0$.

F. Local min / max Values: By the first Derivative test since f' changes from (+) to (−) at 0, $f(0)=0$ is a local maximum



G. Concavity and points of inflection: $f'(x) = \frac{-4x}{(x^2-1)^2}$.

$$f''(x) = \frac{-4(x^2-1)^2 - (-4x)(2)(x^2-1)(2x)}{(x^2-1)^4} = \frac{(x^2-1)\left(-4(x^2-1) + 16x^2\right)}{(x^2-1)^4}$$

$$= \frac{-4x^2 + 4 + 16x^2}{(x^2-1)^3} = \frac{12x^2+4}{(x^2-1)^3}.$$

Notice that $12x^2+4>0$ for $x \in Dom f$.

So, $f''(x)>0 \Leftrightarrow (x^2-1)^3>0 \Leftrightarrow x^2-1>0 \Leftrightarrow (x+1)(x-1)>0$

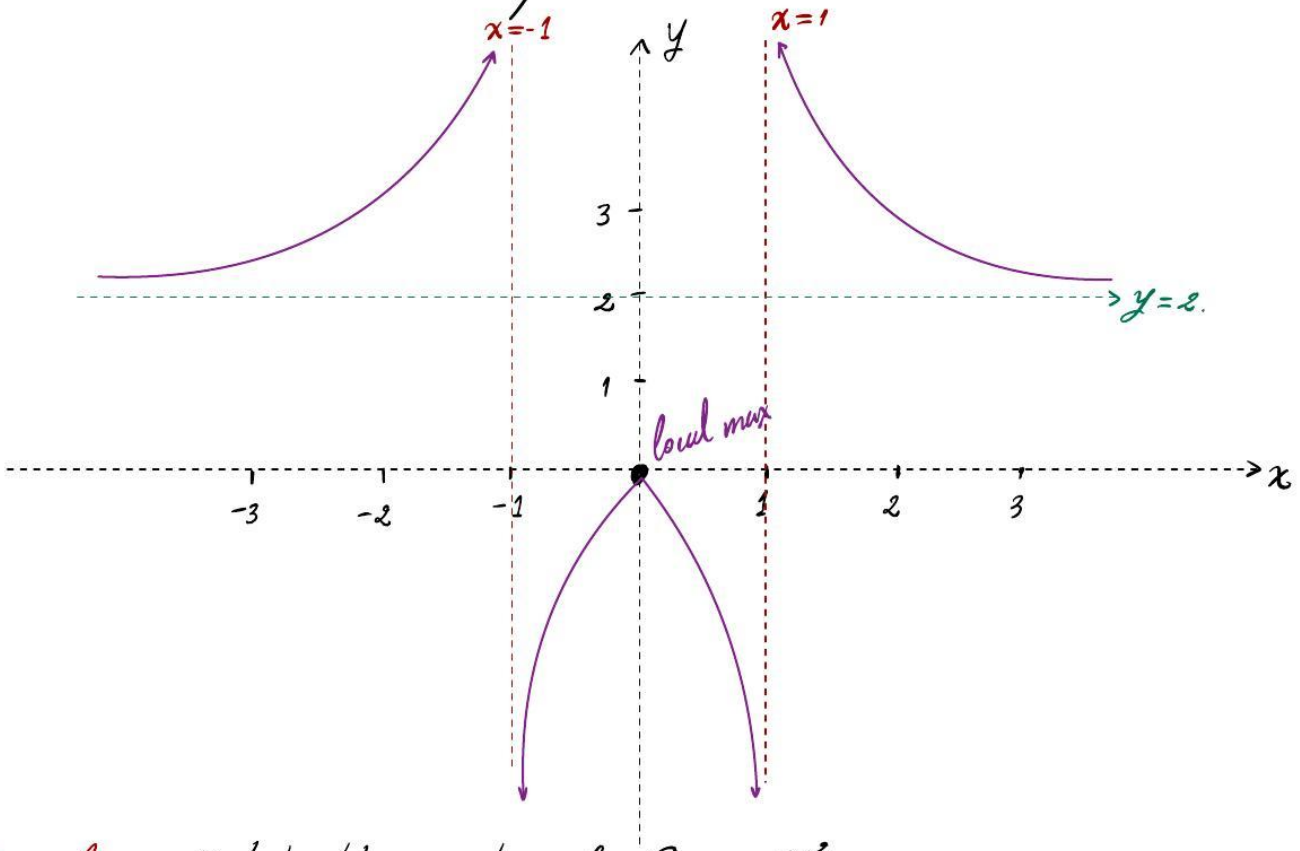


So f is $\cup$ on $(-\infty,-1) \cup (1,\infty)$

f is $\cap$ on $(-1,1)$.

$f''(x) \neq 0$ b/c $12x^2+4>0$ and $f''(x)$ DNE at $x=\pm 1$. but $\pm 1 \notin Dom f$

then there is not inflection points.



Example. Sketch the graph of $f(x)=\dfrac{x^2}{\sqrt{x+1}}$.

Sln. A. $Dom f = (-1,\infty)$.

B. the x- and y- intercepts are both (0,0).

C. Symmetry : None

$$f(-x)=\frac{(-x)^2}{\sqrt{-x+1}}=\frac{x^2}{\sqrt{1-x}} \neq f(x)$$
$$\neq -f(x).$$

D. Asymtotes: Horizontal:

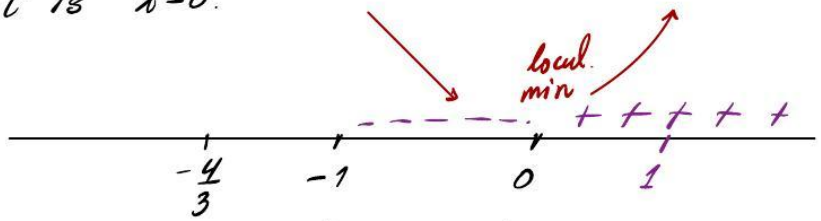
$$\lim_{x\to\infty}\frac{x^2}{\sqrt{x+1}}=\infty$$ there is no H.A.

$\lim_{x \to -1^+} \frac{x^2}{\sqrt{x+1}} = \infty$ So, $x=-1$ is a vertical asymptote.

E. $f'(x) = \frac{2x\sqrt{x+1} - x^2\left(\frac{1}{2\sqrt{x+1}}\right)}{x+1} = \frac{4x(x+1) - x^2}{2\sqrt{x+1}\,(x+1)} = \frac{3x^2+4x}{2\sqrt{x+1}\,(x+1)}$

$f'(x)=0 \iff 3x^2+4x=0 \iff x=0,\ x=-\frac{4}{3} \notin \text{Dom } f$ the only critical point is $x=0$.

$f'(x) > 0$



So f is increasing on $(0,\infty)$ and decreasing on $(-1,0)$.

E. So by the first derivative test $x=0$, $f(0)=0$ is local minimum

F. $f''(x) = \frac{(6x+4)(2(x+1)^{3/2}) - x(3x+4)(3(x+1)^{1/2})}{(2(x+1)^{3/2})^2}$

$= \frac{2(6x+4)(x+1)^{3/2} - 3x(3x+4)(x+1)^{1/2}}{4(x+1)^3}$

$= \frac{(x+1)^{1/2}\left[2(6x+4)(x+1) - 3x(3x+4)\right]}{4(x+1)^3}$

$= \frac{2(6x^2+10x+4) - 9x^2 - 12x}{4(x+1)^{5/2}}$

$= \frac{3x^2+8x+8}{4(x+1)^{5/2}}$.

$3x^2+8x+8=0 \iff x = \frac{-8 \pm \sqrt{8^2-4(3)(8)}}{6} = \frac{-8 \pm \sqrt{64-96}}{6}$

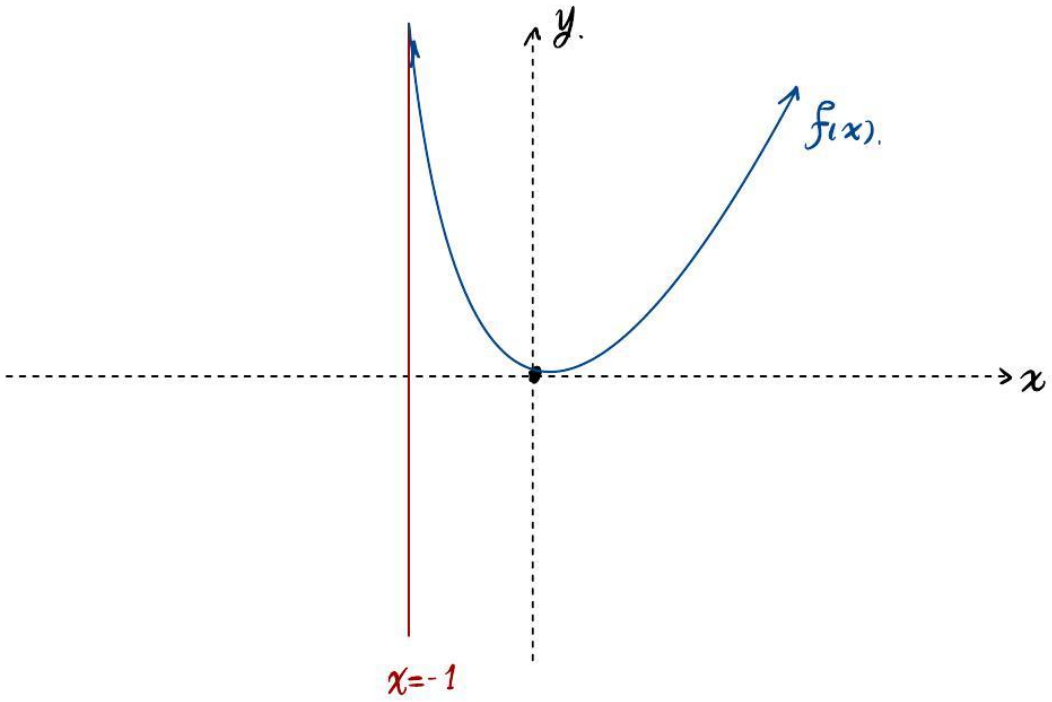
then $3x^2+8x+8>0$ or $3x^2+8x+8<0$ for all $x \in \text{Dom } f$.

$x=1 \in \text{Dom } f$, $3(1)^2+8(1)+8 = 21 > 0$

then $3x^2+8x+8>0$ for all $x \in \text{Dom } f$.

$f''(x) > 0 \Leftrightarrow (x+1)^{5/2} > 0 \Leftrightarrow \sqrt{(x+1)^5} > 0$ then,

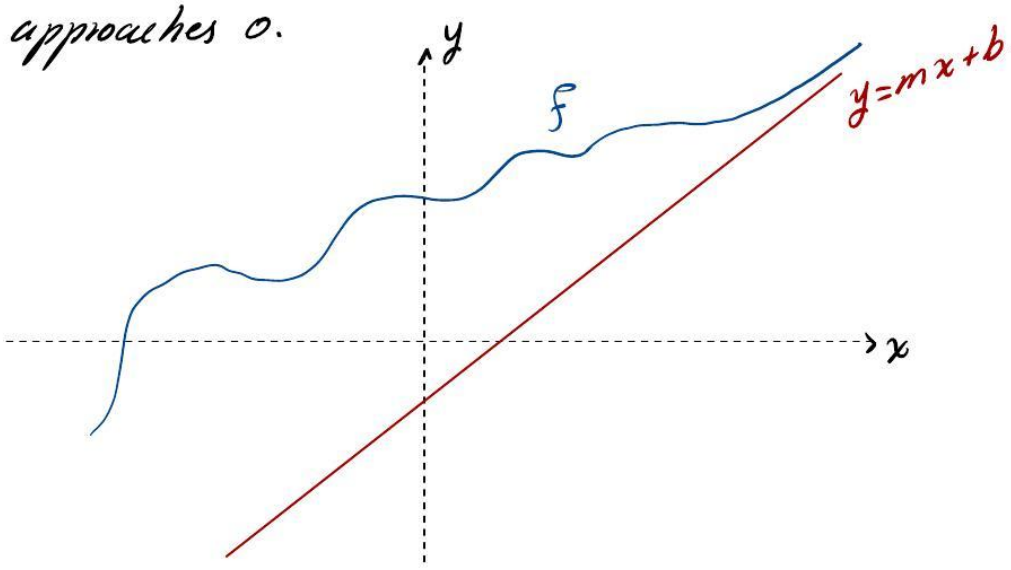
f is $\cup$ on $Dom f = (-1, +\infty)$.



Slant Asymtotes: Some curves have asymtotes that are oblique, that is, neither horizontal nor vertical. if

$$\lim_{x \to \infty} [f(x) - (mx+b)] = 0$$

where $m \neq 0$, then $y = mx+b$ is called a slant asymtote b/c the vertical distance between the curve $y = f(x)$ and the line $y = mx+b$ approaches 0.



Example. Compute slant asymtote of $f(x) = \frac{x^3}{x^2+1}$.

Sln.

$$\begin{array}{r|l} x^3 & x^2+1 \\ \hline -x^3-x & x \\ \hline -x & \end{array}$$

we done if degree here is < degree x^2+1.

then $y=x$ is the possible slant asymtote

$$\lim_{x\to\infty}\left[\frac{x^3}{x^2+1} - x\right] \overset{?}{=} 0$$

$$\lim_{x\to\infty}\left[\frac{x^3-x^3-x}{x^2+1}\right] = \lim_{x\to\infty}\frac{-x}{x^2+1}$$

$$= \lim_{x\to\infty}\frac{-x/x^2}{x^2/x^2 + 1/x^2}$$

$$= \lim_{x\to\infty}\frac{-1/x}{1+1/x^2} = 0.$$

then $y=x$ is a slant asymtote of $y=f(x)$.

- $\text{Dom } f = \mathbb{R}$
- x- and y- intercepts are $(0,0)$.
- $f(-x) = \frac{(-x)^3}{(-x)^2+1} = -\frac{x^3}{x^2+1} = -\frac{x^3}{x^2+1} = -f(x)$ is an odd function.
- No asymtotes.
-

$$f'(x) = \frac{3x^2(x^2+1) - x^3(2x)}{(x^2+1)^2} = \frac{x^4+3x^2}{(x^2+1)^2} = \frac{x^2(x^2+3)}{(x^2+1)^2}$$

f is increasing on $(-\infty, +\infty)$ $x=0$ is the only critical point.

at 0, there is not local min or max By the first Derivative test

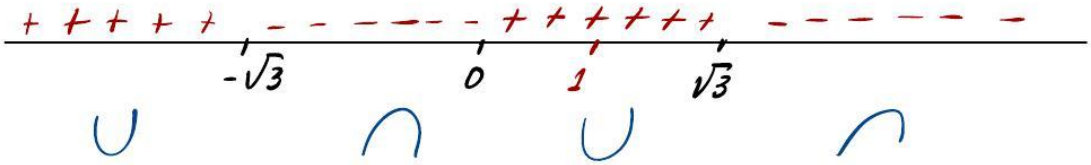
$$f''(x) = \frac{(4x^3+6x)(x^2+1)^2 - (x^4+3x^2)(2)(x^2+1)(2x)}{(x^2+1)^4}$$

$$= \frac{(x^2+1)\left((4x^3+6x)(x^2+1) - 4x(x^4+3x^2)\right)}{(x^2+1)^4}$$

$$= \frac{2x\left((2x^2+3)(x^2+1) - 2x^4 - 6x^2\right)}{(x^2+1)^3}$$

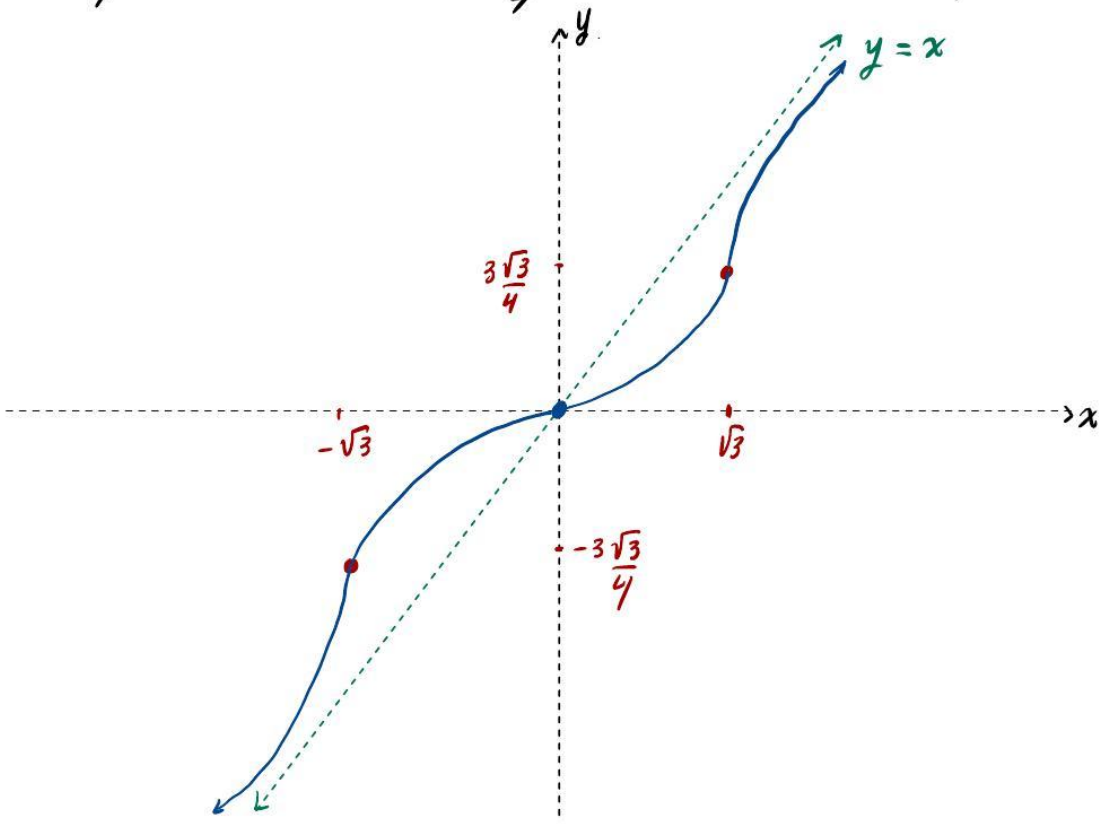
$$= \frac{2x(2x^4+5x^2+3-2x^4-6x^2)}{(x^2+1)^3}$$

$$= \frac{2x(3-x^2)}{(x^2+1)^3} \qquad f''(x)=0 \iff x=0,\ x=\pm\sqrt{3}.$$



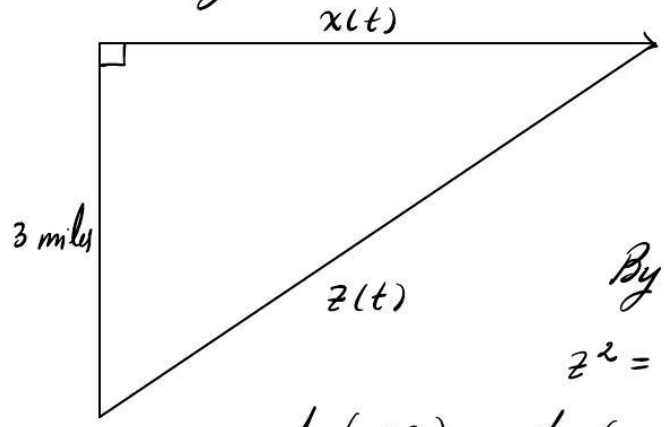
$x=\sqrt{3}$, $f(\sqrt{3}) = \frac{3\sqrt{3}}{4}$; $f(-\sqrt{3}) = -f(\sqrt{3}) = -\frac{3\sqrt{3}}{4}$

$(\sqrt{3}, \frac{3\sqrt{3}}{4})$ and $(-\sqrt{3}, -\frac{3\sqrt{3}}{4})$ are inflection points.



Applications of Implicit Differentiation.

Problem 1 A plane flying horizontally at an altitude of 3 miles and a velocity of 300 mi/hr passes directly over a radar station. Find the rate at which the straight line distance from the plane to the station is increasing when the plane straight line distance of 5 miles away from the station.



$\frac{dx}{dt} = 300 \text{ m/h}$.

$\frac{dz}{dt} = ?$ when $z = 5$ miles.

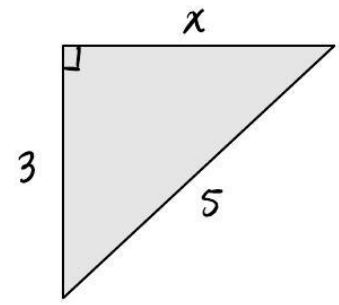
By P-thm.

$$z^2 = x^2 + 3^2$$

$$\frac{d}{dt}\left(z^2\right) = \frac{d}{dt}\left(x^2 + 3^2\right)$$

$$2z \cdot \frac{dz}{dt} = 2x \cdot \frac{dx}{dt} + 0. \quad ; \quad z = 5, \ \frac{dx}{dt} = 300$$

$$\frac{dz}{dt} = \frac{x}{z}\frac{dx}{dt}$$



$5^2 = x^2 + 3^2 \Rightarrow x = \sqrt{25-9} = 4.$

plug in $z = 5$, $x = 4$; $\frac{dz}{dt} = \frac{4}{5}(300) = 240 \text{ mi/hr}$.